

## Program 3 : Code a Dockerized Python Flask or Node.js Application

Step 1 : mkdir program3

cd program3

nano Dockerfile

```
1rv24mc040-haripriya@haripriya-ASUS: ~/Devops_lab/program3
GNU nano 7.2 Dockerfile
FROM python:latest
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY . .
EXPOSE 5000
CMD ["python","app.py"]
```

Step 2 : nano requirements.txt

```
1rv24mc040-haripriya@haripriya-ASUS: ~/Devops_lab/program3
GNU nano 7.2 requirements.txt
Flask==2.3.3
```

Step 3 : nano app.py

```
1rv24mc040-haripriya@haripriya-ASUS: ~/Devops_lab/program3
GNU nano 7.2 app.py
from flask import Flask
app=Flask(__name__)
@app.route("/")
def home():
    return "Hello from Docker Flask"
if __name__ == "__main__":
    app.run(host="0.0.0.0",port=5000)
```

Step 4 : docker build -t program3 .

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops_lab/program3$ docker build -t program3 .
[+] Building 2.4s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 192B                                0.0s
=> [internal] load metadata for docker.io/library/python:latest    2.3s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/python:latest@sha256:1ad1a43b5e2478e6205 0.0s
=> [internal] load build context                                  0.0s
=> => transferring context: 456B                                       0.0s
=> CACHED [2/5] WORKDIR /app                                       0.0s
=> CACHED [3/5] COPY requirements.txt .                             0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt 0.0s
=> [5/5] COPY . .                                                 0.0s
=> exporting to image                                              0.0s
=> => exporting layers                                              0.0s
=> => writing image sha256:f4a880739c448806d3824031c3800d5f450cd7b54ac2d 0.0s
=> => naming to docker.io/library/program3                          0.0s

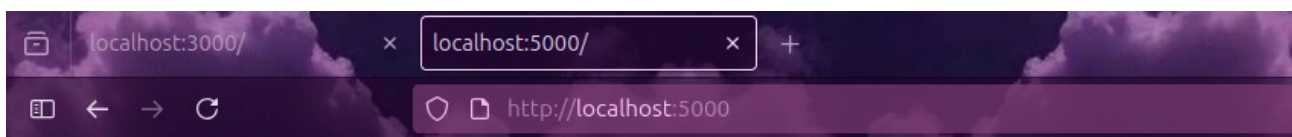
View build details: docker-desktop://dashboard/build/default/default/g0bfx62rjufudwc8p67mja
gdq
```

Step 5 : docker run -p 5000:5000 program3

```
.....
1rv24mc040-haripriya@haripriya-ASUS:~/Devops_lab/program3$ docker run -p 5000:5000 program3
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.18.0.3:5000
Press CTRL+C to quit
172.18.0.1 - - [07/Nov/2025 04:36:44] "GET / HTTP/1.1" 200 -
172.18.0.1 - - [07/Nov/2025 04:36:44] "GET /favicon.ico HTTP/1.1" 404 -
```

Step 6 : Check the output in the browser

<http://localhost:5000>



Hello from Docker Flask